

Homework 1

Uriel Benítez Linear Algebra Sep 17, 2026 GRADE: 21.5/40 (Prof. R. Castellano)

Problem 1

(a)

Final answer: 24

[PROF: -0.5 E1 No work shown; $2u \cdot v$ and the dot product are never computed.]

(b)

$$(1/2) w = [-2, 1, 3]^T$$

$$v + (1/2) w = [1, 4, -2]^T + [-2, 1, 3]^T = [-1, 5, 1]^T$$

$$\|v + (1/2) w\| = (-1)^2 + (5)^2 + (1)^2 = 1 + 25 + 1 = 27$$

[ANSWER: 27]

[PROF: -0.5 E8 The square root is dropped; the norm is $\sqrt{27} = 3\sqrt{3}$, not 27.]

(c)

$$\text{use } \cos \theta = (u \cdot w) / (\|u\| \|w\|).$$

$$u \cdot w = (2)(-4) + (-1)(2) + (3)(6) = -8 - 2 + 18 = 8$$

$$\|u\| = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{4 + 1 + 9} = \sqrt{14}$$

$$\|w\| = \sqrt{(-4)^2 + 2^2 + 6^2} = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$\cos \theta = 8 / (\sqrt{14})(2\sqrt{14}) = 8 / 28 = 2/7$$

$$\theta = \arccos(2/7) = 73.3985\dots^\circ$$

[ANSWER: $\theta \approx 73.40^\circ$]

[PROF: ✓]

[PROF: Problem 1 4/5]

Problem 2

(didn't get to this)

(a) [PROF: 0/1 S6 Not attempted.]

(b) [PROF: 0/1 S6 Not attempted.]

(c) [PROF: 0/1 S6 Not attempted.]

(d) [PROF: 0/1 S6 Not attempted.]

(e) [PROF: 0/1 S6 Not attempted.]

(f) [PROF: 0/1 S6 Not attempted.]

[PROF: Problem 2 0/6]

Problem 3

(a)

Augmented matrix:

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & 1 & | & 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R2 \rightarrow R2 - 2R1: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & 2 & | & -1 \end{bmatrix}$$

$$R3 \rightarrow R3 + R1: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 3 & | & 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$R2 \rightarrow -R2: \begin{bmatrix} 1 & 2 & -1 & | & -1 \end{bmatrix}$$

$$[0 \ 1 \ -3 \mid -4]$$

$$[0 \ 3 \ 1 \mid -2]$$

$$R_3 \rightarrow R_3 - 3R_2: [1 \ 2 \ -1 \mid -1]$$

$$[0 \ 1 \ -3 \mid -4]$$

$$[0 \ 0 \ 10 \mid 10]$$

$$R_3 \rightarrow (1/10)R_3: [1 \ 2 \ -1 \mid -1]$$

$$[0 \ 1 \ -3 \mid -4]$$

$$[0 \ 0 \ 1 \mid 1]$$

Back substitute: $x_3 = 1$, then $x_2 = -4 + 3(1) = -1$, then $x_1 = -1 - 2(-1) + 1 = 2$.

One solution only.

$$\text{[ANSWER: } x = [2, -1, 1]^T \text{]}$$

[PROF: ✓]

(b)

$$[1 \ 2 \ -1 \mid 4]$$

$$[2 \ 5 \ 1 \mid 6]$$

$$[3 \ 7 \ 0 \mid 10]$$

$$R_2 \rightarrow R_2 - 2R_1: [1 \ 2 \ -1 \mid 4]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[3 \ 7 \ 0 \mid 10]$$

$$R_3 \rightarrow R_3 - 3R_1: [1 \ 2 \ -1 \mid 4]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$R_3 \rightarrow R_3 - R_2: [1 \ 2 \ -1 \mid 4]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 0]$$

$$R_1 \rightarrow R_1 - 2R_2: [1 \ 0 \ -7 \mid 8]$$

$$[0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 0]$$

Column 3 has no pivot, so z is free and there are infinitely many solutions.

[ANSWER: infinitely many solutions, z is free]

[PROF: -1 E4 The free variable is named but no parametric general solution is given.]

[PROF: Problem 3 5/6]

Problem 4

(a)

Let x_1 = mass of Alloy 1 used, in kg.

Let x_2 = mass of Alloy 2 used, in kg.

Let x_3 = mass of Alloy 3 used, in kg.

The 100 kg of target alloy holds 36 kg copper, 44 kg zinc, and 20 kg nickel, and metal in = metal out, so:

$$\text{copper: } 0.5x_1 + 0.2x_2 + 0.4x_3 = 36$$

$$\text{zinc: } 0.3x_1 + 0.7x_2 + 0.2x_3 = 44$$

$$\text{nickel: } 0.2x_1 + 0.1x_2 + 0.4x_3 = 20$$

[PROF: ✓]

(c)

Times 10 to kill the decimals:

$$5x_1 + 2x_2 + 4x_3 = 360$$

$$3x_1 + 7x_2 + 2x_3 = 440$$

$$2x_1 + x_2 + 4x_3 = 200$$

$$R_1 \rightarrow R_1 - 2R_3: \quad [1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R_2 \rightarrow R_2 - 3R_1: \quad [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 7 \ 14 \mid 520]$$

$$[2 \ 1 \ 4 \mid 200]$$

$$R_3 \rightarrow R_3 - 2R_1: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 7 \ 14 \mid 520]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_2 \rightarrow (1/7)R_2: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 520/7]$$

$$[0 \ 1 \ 12 \mid 280]$$

$$R_3 \rightarrow R_3 - R_2: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 520/7]$$

$$[0 \ 0 \ 10 \mid 1440/7]$$

$$R_3 \rightarrow (1/10)R_3: [1 \ 0 \ -4 \mid -40]$$

$$[0 \ 1 \ 2 \mid 520/7]$$

$$[0 \ 0 \ 1 \mid 144/7]$$

$$x_3 = 144/7 \approx 20.57$$

$$x_2 = 520/7 - 2(144/7) = 232/7 \approx 33.14$$

$$x_1 = -40 + 4(144/7) = -40 + 576/7 = 296/7 \approx 42.29$$

[ANSWER: 42.29 kg of Alloy 1, 33.14 kg of Alloy 2, 20.57 kg of Alloy 3]

[PROF: -1 E8 In $R_2 \rightarrow R_2 - 3R_1$ the constant is $440 - 3(-40) = 560$, not 520; the slip is carried consistently.]

(b) [PROF: 0/1 S6 Not attempted.]

[PROF: Problem 4 6/8]

Problem 5

(a)

$$R_2 \rightarrow R_2 - 2R_1: \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 3 & 6 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 3 & 6 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 3 & 6 & 4 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 3R_1: \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2: \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - R_2: \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[\text{ANSWER: RREF}(A) = [120; 001; 000]]$$

[PROF: -1 E8 In $R_3 \rightarrow R_3 - 3R_1$ the third entry is $4 - 3(1) = 1$, not 2; the slip is carried consistently.]

(b)

$$\text{rank}(A) = 2$$

[PROF: -0.5 E7 The rank is stated with no pivot-count justification.]

(c)

0 solutions: possible.

1 solution: impossible.

Infinitely many: possible.

[PROF: -4 E7 Three bare verdicts with no reasons; an unjustified claim about the number of solutions earns no credit.]

[PROF: Problem 5 1.5/71]

Problem 6

(a)

Put v_1, v_2, v_3 in as columns and row reduce:

$$\begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1: \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2: \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Two pivots but three vectors, so the set is linearly dependent.

The relation is $v_3 = v_2 - 2v_1$, since $[2, 1, 3]^T - [2, 0, 4]^T = [0, 1, -1]^T$.

v_1 and v_2 are not multiples of each other, so they span a plane through the origin.

[ANSWER: dependent; span is a plane in $\mathbb{R}^3$]

[PROF: ✓]

(b)

I'll use the determinant this time. Columns v_1, v_2, v_3 :

$$\begin{vmatrix} 1 & 0 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 1 & 0 \end{vmatrix}$$

$$\begin{vmatrix} 0 & 1 & 1 \end{vmatrix}$$

Expand across the top row:

$$\begin{aligned}\det &= 1(1 \cdot 1 - 0 \cdot 1) - 0(1 \cdot 1 - 0 \cdot 0) + 1(0 \cdot 1 - 1 \cdot 1) \\ &= 1(1) - 0 + 1(-1) \\ &= 1 - 1 = 0\end{aligned}$$

The determinant is 0, so the columns are linearly dependent.

Since they are dependent but not all multiples of one vector, the span is a plane.

[ANSWER: dependent; span is a plane in $\mathbb{R}^3$]

[PROF: -1 E8 The third minor is $\det[[1,1],[0,1]] = 1$, so $\det = 2$, not 0.; -2 S3 The slip reverses the conclusion, so the independence verdict and the geometric description are both lost.]

[PROF: Problem 6 5/8]

Grader summary

Problem 1: 4/5

Problem 2: 0/6

Problem 3: 5/6

Problem 4: 6/8

Problem 5: 1.5/7

Problem 6: 5/8

Total: 21.5/40

Problem 2 is blank and Problem 4(b) was never written; that is 7 points you did not attempt. Where you do show the elimination the work is clean and labeled, but you state verdicts without reasons in 5(b) and 5(c), and E7 gives no credit for an unjustified claim about rank or the number of solutions. Check each row operation as you write it: the slips in 4(c), 5(a), and 6(b) cost 3 points directly, and the one in 6(b) flipped your conclusion for 2 more.